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Reaction-Based Quantification of Surface Brønsted Acid Site Densities in Large-Pore Zeolites and Their Consequences for Bulky Molecule Transformations

Cole W. Hullfish | Jun Zhi Tan | Carson J. Repins | Michele L. Sarazen

Department of Chemical and Biological Engineering, Princeton University, Princeton, New Jersey, USA

Correspondence: Michele L. Sarazen (msarazen@princeton.edu)

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ABSTRACT

This study extracts densities of surface Brønsted acid sites (SBAS) in large-pore zeolites via reactions of bulky micropore-excluded molecules. Self-etherification of 1-pyrenemethanol, a polycyclic primary alcohol not accessible to protons within medium- or large-pore zeolites, was measured on a suite of zeolites. A linear correlation was established between extracted zero-order rate constants and independently measured SBAS densities in a series of MFI zeolites and was extended to large-pore frameworks. Parity among MFI, MOR, BEA, and FAU zeolites in SBAS-catalyzed Hofmann elimination reactions indicated accurate SBAS quantification and was consistent with Brønsted acid strength that is independent of pore topology and location (i.e., internal, external, T-site) in the absence of pore confinement effects. SBAS densities in the large-pore frameworks were used to inform reaction mechanisms under severe internal diffusion limitation, as in polyolefin upcycling. Polyethylene (PE) catalytic cracking rates were found to be commensurate with increasing SBAS densities, although ingress of SBAS-cleaved species still contributed significantly to rates. Extensively, SBAS densities—and the universal methodology employed to quantify them, herein—can be widely used to inform a broad range of catalytic systems involving diffusion-limited or -prohibited molecules.

1 | Introduction

Pore architectures of zeolite catalysts define confining environments around Brønsted acid sites (BAS) that dictate stabilization of guest molecules and kinetically relevant transition state species within micropores of molecular dimension [1–6]. These confinement effects can heavily influence zeolite-catalyzed reactions as has been observed in comparison of rates and selectivities that depend on zeolite pore size, dimensionality, geometry, and even placement within specific channels or intersections [7–10]. BAS can also reside on external surfaces (for example, SBAS), but their contribution to overall rates is often inconsequential due to BAS that typically outnumber SBAS by orders of magnitude and absence of confinement effects in SBAS that results in an effective lower reactivity relative to microporous BAS. However, catalytic effects of SBAS can be non-negligible in the presence of internal mass transfer limitation or when reactant molecules (or species undergoing secondary reactions) are unable to access internal micropores [11–13].

In such cases, SBAS can be accounted for through titrations with molecules such as 2,6-di-*tert*-butylpyridine (DTBP) that are diffusion-prohibited from accessing internal micropores in small- and medium-pore zeolites. These titrations can be performed in situ during reaction through cofeeds or used in conjunction with techniques such as Fourier-transform infrared spectroscopy or temperature-programmed desorption [8, 13–15]. These methods work reasonably well for small- and medium-pore zeolites, but are unviable for large-pore frameworks since DTBP can access internal micropores in zeolites like BEA and FAU [16, 17]. Thus, large-pore zeolites necessitate the use of bulkier molecules such that only SBAS are probed. 2,4,6-tri-*tert*-butylpyridine has been explored as an alternative titrant to DTBP to measure SBAS densities in large-pore zeolites [17–21], but its solid form at room temperature and the need for extinction coefficients can pose difficulties for dosing and spectrometric quantification. In contrast to titration, reaction-based techniques avoid such complications and enable quantification of SBAS densities that may fall beneath instrument sensitivity thresholds [14]. Work by Pereira and Gorte revealed that internal and external BAS in FAU zeolites could be distinguished through thermogravimetric Hofmann elimination reactions with 2,2-diphenylethylamine (DPEA) [22]. Similarly, Wu, Wight, and Davis exploited the inability of 1-pyrenemethanol (PM) to diffuse into FAU micropores to confirm the absence of perruthenate on external surfaces (and therefore, predominant perruthenate encapsulation within FAU supercages) [23]. Reactions such as these highlight the potential of chemical transformations of bulky molecules on external zeolite surfaces to quantify SBAS densities in large-pore frameworks otherwise impracticable via titration or impossible using conventional probe molecules like DTBP.

This study presents a method to quantitatively measure SBAS densities in large-pore zeolites that takes advantage of size exclusion principles to relegate catalytic turnovers solely to SBAS. A suite of MFI catalysts with similar pore topologies, but SBAS densities that span orders of magnitude was synthesized via post-synthetic steaming and ammonium hexafluorosilicate treatments and utilized as a reference to benchmark reactivity, and therefore SBAS density for large-pore frameworks. Specifically, this series of catalysts is used to construct a linear correlation between zero-order PM self-etherification rate constant and SBAS density, indicating that PM self-etherification rates on zeolite surfaces solely depend on the number of accessible SBAS. Zero-order self-etherification rate constants are measured for MOR, BEA, and FAU zeolites and mapped to SBAS densities via the linear correlation. These SBAS densities are consistent with measured alkene production in DPEA Hofmann elimination reactions, suggesting that SBAS values extracted via this technique are quantitative.

These findings indicate apparent parity between reaction rates on external surfaces and SBAS counts that may experimentally evidence equal reactivity of BAS regardless of zeolite framework when protons reside on extracrystalline surfaces. Further, SBAS counts for BEA, MOR, and FAU were ultimately able to inform reactivity trends for reactions under extreme diffusion limitation such as polyethylene (PE) catalytic cracking, where it was revealed that access to internal protons requires initial activation on SBAS, and that these activation events interplay with diffusion of cleaved chains to dictate solid conversion rates. Insights from this work emphasize the enhanced role of SBAS in sub-unity effectiveness factor regimes, a phenomenon that can be quantitatively accounted for by the methodology developed herein.

2 | Methods

2.1 | Catalyst Synthesis and Modification

Zeolite powders were purchased from Zeolyst International and include the NH_4^+ forms of MFI with nominal Si/Al = 25, 40, and 140 (CBV 5524G, CBV 8014, CBV 28014), MOR with Si/Al = 10 (CBV 21A), FAU with Si/Al = 15 and 40 (CBV 720, CBV 780), and BEA with Si/Al = 12.5 (CP814E). Zeolites were calcined in a tube furnace under flowing air (Airgas, zero grade; >100 sccm g_{cat}^{-1}) at 823 K (2 K min^{-1}) for 8 h to yield the respective proton forms denoted H-ZEO-X, where ZEO indicates the framework type and *X* indicates Si/Al. A silicalite-1 material was synthesized utilizing methods previously reported [24] and characterized in an earlier work [25].

Parent zeolites subjected to mild steaming treatments were heated in the tube furnace with 1″ tube ends open to atmosphere to allow exposure to water vapor natively present in air to induce slight reductions in Brønsted acid site (BAS) densities, primarily through conversion to extraframework aluminum (EFAl), without causing extensive dealumination. H-MFI-25 and H-MFI-40 zeolites were ramped to 823 K (2 K min^{-1}), held for 8 h, and cooled to ambient to yield EFAl-H-MFI-X. A published procedure was followed to subject parent zeolites to ammonium hexafluorosilicate (AHFS) treatments [14]; a 0.016 M solution of AHFS (Thermo Fisher Scientific, Puratronic™, 99.999%) in deionized H_2O (18.2 MΩ·cm) in a capped polytetrafluoroethylene Parr liner was first preheated to 353 K in an oil bath under stirring. H-MFI-25 or H-MFI-40 was added (0.025 g_{cat} cm^{-3} AHFS solution) and allowed to stir for 5 h, after which the solids were washed with deionized H_2O via successive centrifugation (500 cm^3 g_{cat}^{-1}). After drying at 353 K for 8 h in a convection oven (Yamato, DKN402C), the AHFS-treated zeolites were calcined identically to the parent NH_4^+ zeolites to yield AHFS-H-MFI-X.

An aluminosilicate MCM-41 catalyst was obtained from Sigma-Aldrich (Product No. 643653). The as-received powder was ion-exchanged in aqueous NH_4NO_3 solution as described above and calcined identically to the zeolite samples. This catalyst is denoted Al-MCM-41.

Sodium form catalysts were synthesized from proton form parents via ion exchange in an aqueous 1 M NaCl solution (EMD Chemicals, 99.0%; 150 cm^3 g_{cat}^{-1}) at ambient temperature for 24 h [26]. The solids were washed with deionized H_2O via successive centrifugation (300 cm^3 g_{cat}^{-1}), dried in the convection oven at 353 K for 8 h, and calcined as stated previously.

2.2 | Material Characterization

X-ray diffractometry (XRD) was performed by a Bruker D8 Discover x-ray diffractometer with Cu K_α radiation (λ = 1.5418 Å) to confirm zeolite crystallinity. A small amount of vacuum grease was applied to a glass microscope slide, onto which ~10 mg of zeolite was loaded. The catalyst-glass slide ensemble was mounted vertically for XRD analysis. Transmission electron microscopy (TEM) was conducted using a Talos F200X S/TEM microscope at 200 kV accelerating voltage. To prepare catalysts for TEM imaging, ~1 mg of catalyst and ~1.5 cm^3 of acetone were transferred to a 3.7 cm^3 borosilicate scintillation vial and sonicated for ~0.25 h to disperse the crystallites. Following sonication, 0.01–0.03 cm^3 of the suspension was transferred via micropipette to a wafer of copper mesh-backed carbon film and dried under ambient conditions for ~0.5 h. Catalyst Si/Al ratios were determined by energy-dispersive x-ray (EDX) spectroscopy using a Quanta 200 FEG environmental-scanning electron microscope equipped with an EDX accessory. N_2 physisorption at 77 K was performed with a Micromeritics 3Flex instrument to assess catalyst porosity metrics. Prior to sample analysis, catalysts were degassed at >383 K for 24 h under vacuum (~125 Torr). Brunauer-Emmett-Teller surface areas (S_{BET}) were calculated adhering to the Rouquerol criteria using the BET Surface Identification software [27, 28]. Mesopore surface areas and volumes were determined via the Barrett-Joyner-Halenda (BJH) model using the adsorption isotherm, and micropore volumes were determined via *t*-plot method.

2.3 | Liquid-Phase Alkylation and Self-Etherification Reactions

Alkylation reactions between 1,3,5-trimethylbenzene (TMB) and dibenzyl ether (DBE) were performed following published protocol to extract SBAS densities of parent and modified MFI zeolites [14]. A 50 cm^3 three-neck round-bottom flask was equipped with a stopper, reflux condenser, and rubber septum, and 100 mg of catalyst was added. The apparatus was transferred to a 393 K oil bath and dehydrated for 1 h with venting enabled via a 20 G needle piercing the septum. The needle was then removed, the bath was cooled to 363 K, and 5 cm^3 of TMB (Thermo Scientific, 99%, extra pure) was dosed to the catalyst. The catalyst-TMB mixture equilibrated for 0.5 h under stirring and reflux after which 0.2 cm^3 of DBE (Sigma–Aldrich, 98%), was injected to initiate the reaction. Periodic 0.15 cm^3 aliquots were extracted via the stopper port, immediately filtered through cotton-plugged pipettes, and transferred to $CDCl_3$ (Cambridge Isotope Laboratories, 99.8% D) in a separate scintillation vial for NMR analysis. NMR sample tubes consisted of 0.570 cm^3 $CDCl_3$, 0.025 cm^3 filtered aliquot, and 0.005 cm^3 of an internal standard comprised of 50 mM *N,N*-dimethylformamide (Fisher Chemical, 99.8%) in TMB. Sample compositions were quantified via ^{1}H NMR spectroscopy using a Bruker 500 MHz spectrometer.

1-pyrenemethanol (PM) self-etherification reactions were similarly performed in 50 cm^3 three-neck round-bottom flasks with the same attachments as TMB-DBE alkylation reactions. The process for loading and dehydrating catalysts is also identical, although a lower amount of catalyst was used (3–50 mg). Following catalyst dehydration, 7 cm^3 TMB was added and stirred for 0.5 h at 363 K. During this equilibration, 150 mg PM (TCI, ≥98.0%) was solvated in 4.5 cm^3 TMB via stirring at 363 K in a separate oil bath. To initiate the PM self-etherification reaction, 3 cm^3 of the heated PM solution was transferred via needle and syringe to the round-bottom flask containing the catalyst-TMB mixture to yield a reactant mixture containing 100 mg PM in 10 cm^3 TMB. Reactions times were held to ≤5 min to minimize deactivation with exception to H-MFI-140, which had a significantly low SBAS density. Etherification rates were measured by linear regression of temporal DPE concentration profiles. To extract and quench a high volume of aliquots during the short reaction period, 0.2 cm^3 aliquots were pipetted, filtered, and immediately transferred to prepared $CDCl_3$ solutions. NMR sample tubes consisted of 0.545 cm^3 $CDCl_3$, 0.050 cm^3 aliquot, and 0.005 cm^3 of the same internal standard as TMB-DBE alkylation reactions. Sample compositions were quantified via ^{1}H NMR spectroscopy using a Bruker 500 MHz spectrometer. ^{1}H NMR chemical shifts of the product di-(1-pyrenylmethyl)ether (DPE) were consistent with published literature (δ: 5.36 ppm (s, 4 H, CH_2), δ: 7.8–8.5 ppm (m, 18 H, aromatic); Figure S8) [29].

2.4 | Hofmann Elimination Reactions

Temperature-programmed Hofmann elimination reactions with n-propylamine (NPA) were utilized to determine BAS densities following a procedure shown to give values consistent with reported BAS values for Zeolyst CBV 8014 (H-MFI-40) as determined by ammonia temperature-programmed desorption (TPD), isopropylamine-TPD, and pyridine-IR [30–32]. NPA (Sigma-Aldrich, ≥99%) was dosed to aluminosilicate catalysts (~0.03 cm^3 g_{cat}^{-1}) in a scintillation vial at ambient temperature and allowed to evaporate overnight to saturate BAS with NPA. Dry NPA-dosed catalysts were loaded (~10 mg) into ceramic pans for thermogravimetric analysis (TGA; PerkinElmer, TGA8000). Samples were first heated under 30 sccm Ar flow (Airgas, UHP 99.999%) to 423 K (10 K min^{-1}) and held for 2 h to desorb H_2O and excess weakly adsorbed NPA. Samples were then heated to 673 K (10 K min^{-1}) and held for 0.5 h to perform the Hofmann elimination reaction. Propene formation was monitored via TGA-mass spectrometry (MS; PerkinElmer, Clarus SQ8T) to deconvolute mass losses from residual NPA desorption and Hofmann elimination turnovers. This region typically fell between 598 K and the end of the 673 K hold for aluminosilicates utilized in this study under the described conditions. BAS densities were determined assuming a 1:1 stoichiometry between propene and BAS.

Prior to 2,2-diphenylethylamine (DPEA) Hofmann elimination reactions, catalysts were dosed ex situ with a solution of 70 mg DPEA (ChemScene, 99.65%) dissolved in 10 cm^3 diethyl ether (Alfa Aesar, 99+%, inhibitor free, spectrophotometric grade) in a volumetric flask. Target amounts of DPEA solution were added via micropipette at ambient temperature to ~40 mg catalyst to ensure >20 times excess of DPEA relative to SBAS counts calculated by TMB-DBE alkylation or PM self-etherification reactions. The DPEA-dosed catalysts were dried in a fume hood overnight at ambient temperature, washed with 150 cm^3 diethyl ether via three rounds of centrifugation to remove excess DPEA, and again dried in the fume hood overnight at ambient temperature. To perform DPEA Hofmann elimination reactions, ~15 mg of dried DPEA-dosed catalysterews loaded onto ceramic pans for thermogravimetric analysis. The samples were first ramped under 30 sccm Ar flow to 423 K (10 K min^{-1}) and held for 2 h to remove excess H_2O and diethyl ether prior to ramping again to 673 K (10 K min^{-1}) for a 3 h hold to perform the Hofmann elimination reaction. Following this step, the carrier gas was changed to 40 sccm air (Airgas, Ultra Zero), ramped to 923 K (10 K min^{-1}), and held for 1 h to calcine off organics and obtain the catalyst mass. From TGA-MS analysis, the mass of DPEA converted to the associated alkene via Brønsted acid-catalyzed Hofmann elimination was taken to be the mass loss from ~543 K through the end of the 623 K hold (often appearing as a separate peak in the derivative weight loss curve) for aluminosilicates utilized in this study under the described conditions.

2.5 | Polyethylene Catalytic Cracking Reactions via Large-Pore Zeolites

PE catalytic cracking reactions were performed by proton forms of large-pore zeolites in the TGA using commercial PE (Sigma-Aldrich, 4 kDa) in a 5:1 PE to catalyst ratio by mass. The PE and catalyst mixture was loaded into a ceramic pan (10 mg total) and heated under 30 sccm Ar flow to 423 K (10 K min^{-1}) and held for 2 h to dehydrate the catalyst. A second ramp brought the reactant mixture to the reaction temperature of 523 K (10 K min^{-1}) for 5 h prior to further heating to 873 K to remove unreacted PE. The carrier gas was then switched to 30 sccm air for 1 h at this temperature to burn off residual organics and obtain the catalyst mass. Reported solid conversion rates are all at 5% conversion by mass.

3 | Results

3.1 | Assessment of Crystallinity and Porosity of Parent and Modified Zeolites

MFI zeolites in their parent, EFAl, AHFS, and hierarchical forms constituted a suite of materials of constant pore limiting diameter but varying Al contents and BAS densities. The steaming and AHFS treatments performed on parent H-MFI-25 and H-MFI-40 zeolites did not result in decreases in crystallinity in the resultant modified materials as determined by XRD (Figure 1). Although long-range order was not affected by these treatments, the modified catalysts exhibited measurable differences in Si/Al ratio, Al speciation, and porosity metrics relative to parent forms (Table 1).

Similar Si/Al ratios of EFAl-H-MFI-25 and EFAl-H-MFI-40 (31 and 37) compared to their parent counterparts (31 and 41) indicated that the steaming was not harsh enough to induce extensive dealumination. Zeolite pore structures were also largely unaffected, as micropore volumes, mesopore surface areas and volumes, and total S_{BET} remained approximately constant. Despite these similarities, the mild steaming procedures were strong enough to convert some framework aluminum (in the form of BAS) to EFAl species, as BAS densities for H-MFI-25 and H-MFI-40 decreased by 11% and 8%, respectively. While these deviations to total BAS density are relatively small, SBAS densities in both EFAl-H-MFI-25 and EFAl-H-MFI-40 decreased to a significant degree resultant of the steaming as further discussed in Section 3.2.

In contrast to mild steaming, AHFS treatments did dealuminate H-MFI-25 and H-MFI-40 zeolites as indicated by increases in Si/Al from 31 to 36 in AHFS-H-MFI-25 and from 41 to 62 in AHFS-H-MFI-40. Similarly, total BAS densities decreased by 18% and 28% for H-MFI-25 and H-MFI-40. These losses in total aluminum and BAS are consistent with comparable AHFS procedures that have been shown to leach both framework aluminum and EFAl species in MFI zeolites [14, 33]. However, while some studies report negligible changes to mesopore surface areas and volumes with AHFS treatments [34], AHFS-H-MFI-25 and AHFS-H-MFI-40 exhibited slight reductions to V_{micro} (-0.02 cm^3 g_{cat}^{-1}) with the former also decreasing in S_{meso} (-31%) and V_{meso} (-14%). These pore volume losses were consistent with reported annealing-type behavior involving Si insertion from AHFS into vacancy defects [35, 36]. In contrast, S_{meso} and V_{meso} increased (+16%, +22%) in AHFS-H-MFI-40 relative to the parent H-MFI-40. This increase in mesoporosity was primarily in the 3–10 nm range as indicated by BJH adsorption pore size distributions, whereas AHFS-H-MFI-25 underwent a general decrease in mesopore volume across all pore diameters (Figure 2).

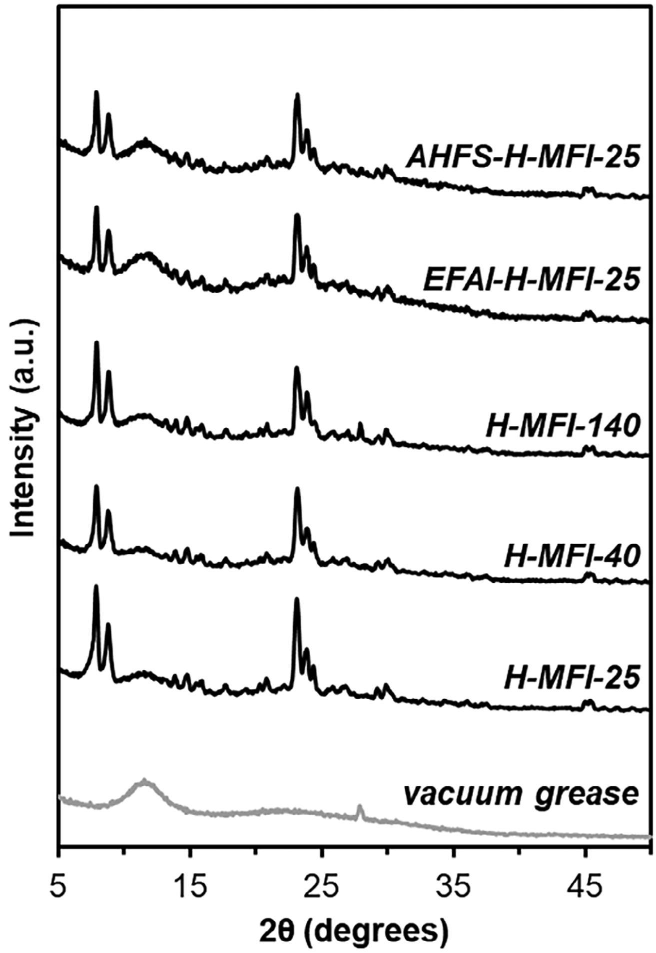


FIGURE 1 | X-ray diffractograms of parent MFI zeolites and MFI zeolites modified by mild steaming and AHFS treatments.

Notably, the differences in mesoporosity between parent and AHFS-MFI zeolites were small relative to typical desilication treatments designed to intentionally incorporate mesopores. Mild desilication treatments (0.2 M NaOH, 333-338 K, 0.5–1 h) can yield ~1.5-5.5 times increases in S_{meso} and V_{meso} for MFI zeolites [30, 37–40], whereas only a ~1.1 times difference was observed in AHFS-H-MFI-25 and AHFS-H-MFI-40 relative to their respective parents. Further, in contrast to desilication, the AHFS treatments resulted in increased Si/Al ratio and decreased total BAS density. Thus, the slight increases in S_{meso} and V_{meso} for AHFS-H-MFI-40 cannot be resultant of desilication-type mechanisms

TABLE 1 | Site and textural properties of the suite of parent and modified aluminosilicate catalysts.

Catalyst	Si/Al[a]	BAS Density[b] (mmol H^+ g_{cat}^{-1})	S_{BET} (m^2 g_{cat}^{-1})	S_{meso}[c] (m^2 g_{cat}^{-1})	V_{micro}[d] (cm^3 g_{cat}^{-1})	V_{meso}[c] (cm^3 g_{cat}^{-1})
H-MFI-25	31	0.45	430	45	0.12	0.07
H-MFI-40	41	0.36	450	75	0.13	0.09
H-MFI-140	142	0.05	390	41	0.11	0.04
EFAl-H-MFI-25	31	0.40	400	32	0.11	0.06
EFAl-H-MFI-40	37	0.33	430	67	0.11	0.08
AHFS-H-MFI-25	36	0.37	410	31	0.10	0.06
AHFS-H-MFI-40	62	0.26	470	87	0.11	0.11
Al-MCM-41	12	0.11[e]	800	480	0.00	0.81
H-MOR-10	9	0.79	460	18	0.17	0.09
H-BEA-12.5	15	0.75	570	230	0.16	0.78
H-FAU-15	17	0.60	840	100	0.26	0.18
H-FAU-40	66	0.21	830	110	0.24	0.20

[a]Determined by EDX.
[b]Determined by NPA temperature-programmed reaction.
[c]Determined by BJH analysis method.
[d]Determined by *t*-plot analysis method.
[e]Determined by TMB-DBE alkylation.

for mesopore formation and instead suggest dealumination and restructuring of inherent mesopores.

Taken together, these parent and modified MFI catalysts comprise a material suite with one shared zeolite framework identity, but varying aluminum contents, aluminum speciation, and BAS densities. The diverse range of catalyst site properties while maintaining similar micropore accessibility makes this set of materials appropriate for assessment of external site densities in reactions utilizing micropore-occluded molecules in the following sections.

3.2 | Determination of SBAS Densities in MFI Zeolites via TMB-DBE Alkylation

Surface Brønsted acid site (SBAS) quantification in zeolites typically involves use of reactants or titrants with molecular diameters that exceed pore limiting diameters of the micropores. For MFI zeolite, Friedel-Crafts alkylation of 1,3,5-trimethylbenzene (TMB) with dibenzyl ether (DBE; Scheme 1a) exclusively occurs on external surfaces because TMB is diffusion-prohibited from accessing internal micropores [14, 40–43]; hence, this reaction is an appropriate probe for SBAS density.

Ezenwa et al. extracted total aluminum-normalized zero-order TMB-DBE alkylation rate constants (k_{TM2B}) for MFI zeolites and plotted the values against total aluminum-normalized SBAS densities obtained via 2,6-di-*tert*-butylpyridine-TPD and obtained a linear correlation [14]. This trend being linear demonstrated that within MFI zeolites (or extensively, any singular small- or medium-pore zeolite framework), measured TMB-DBE alkylation rates are solely a function of the number of isolated BAS accessible to the reaction. Zero-order TMB-DBE alkylation rate constants were measured and mapped to SBAS densities utilizing this linear correlation to determine SBAS densities for the suite of parent and modified MFI zeolites (Figure 3).

Measured k_{TM2B} values and extracted SBAS densities for parent and modified MFI zeolites are given in Table 2. Notably, the SBAS density calculated for H-MFI-40 (0.018 ± 0.001 mmol H^+ g_{cat}^{-1}) was similar to that of the same sample (Zeolyst CBV 8014) utilized by Ezenwa et al. (0.012 ± 0.003 mmol H^+ g_{cat}^{-1}), strongly suggesting that SBAS values obtained via this alkylation reaction were consistent with values obtained via titration-based methods with 2,6-di-*tert*-buytlpyridine. Among the parent zeolites, H-MFI-40 had the highest mass-normalized SBAS density despite a higher relative aluminum content and total BAS density in H-MFI-25 (Table 1). Additionally, H-MFI-40 had nominally larger crystallite diameters than H-MFI-25 (390 ± 210 nm versus 180 ± 100 nm; Figure S3), which would result in a net lower SBAS density relative to H-MFI-25 assuming a random distribution of framework aluminum. However, higher measured SBAS densities in H-MFI-40 indicated Brønsted acid siting is not necessarily governed by bulk physiochemical properties and depends on specific synthesis conditions. Accordingly, combinations of total aluminum and crystallite diameter cannot be used as a reliable proxy for SBAS density as is sometimes invoked.

In addition to surface protons, there could also be an impact of (minor) inherent mesoporosity in H-MFI-40 that enabled alkylation turnovers on BAS accessible via mesopore channels. In this context, SBAS densities measured by TMB-DBE alkylation represented BAS counts on external crystallite surfaces or accessible via mesopores inherent to the zeolites, the latter of which effectively function as surface protons due to absence of pore confinement effects (or similar extents of slight van der Waals interactions). However, SBAS quantification obtained through TMB-DBE alkylation reactions is not dissimilar from

TABLE 2 | Measured aluminum-normalized zero-order TMB-DBE alkylation rate constants and corresponding SBAS densities for parent and modified MFI zeolites. Values in parentheses represent 95% confidence interval uncertainties.

Catalyst	k_{TM2B} (mmol TM2B (mol Al_{tot})$^{-1}$ s^{-1})		SBAS density (mmol H^+ (mol Al_{tot})$^{-1}$)		SBAS density (mmol H^+_{ext} g_{cat}^{-1})	
H-MFI-25	0.033	(0.003)	7.4	(0.9)	0.0039	(0.0004)
H-MFI-40	0.207	(0.003)	46	(3)	0.018	(0.001)
H-MFI-140	0.002	(0.001)	0.4	(0.3)	0.00005	(0.00003)
EFAl-H-MFI-25	0.0027	(0.0001)	0.60	(0.05)	0.00031	(0.00002)
EFAl-H-MFI-40	0.089	(0.002)	20	(1)	0.0087	(0.0006)
AHFS-H-MFI-25	0.019	(0.001)	4.2	(0.3)	0.0019	(0.0001)
AHFS-H-MFI-40	0.53	(0.01)	120	(10)	0.031	(0.002)

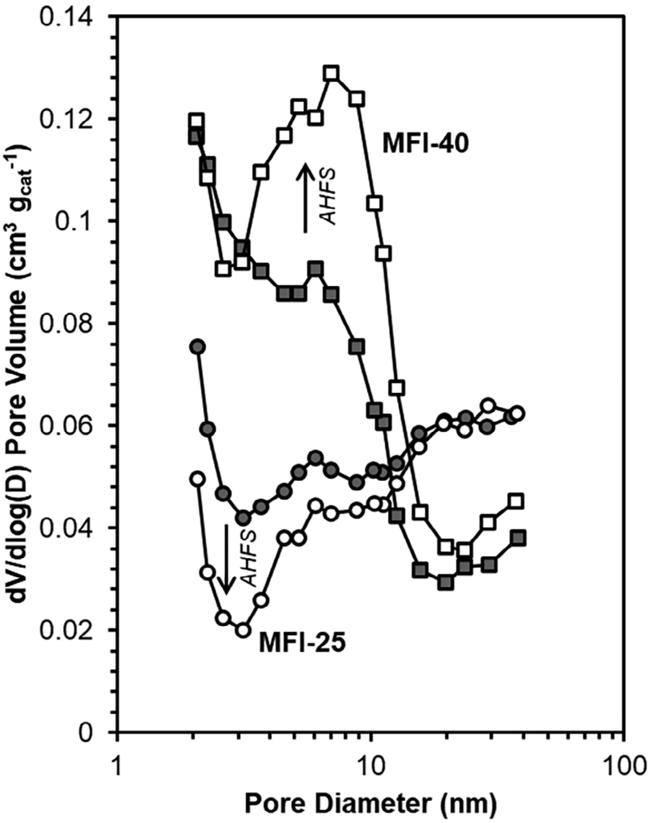


FIGURE 2 | BJH adsorption pore size distributions for MFI-25 (circles) and MFI-40 (squares) for the parent (gray) and AHFS-treated (white) analogs.

commonly employed titration-based methods involving bulky molecules, which also count BAS accessible by mesopores [17] and on external surfaces since such molecules are also not diffusion-prohibited from (non-occluded) mesopores.

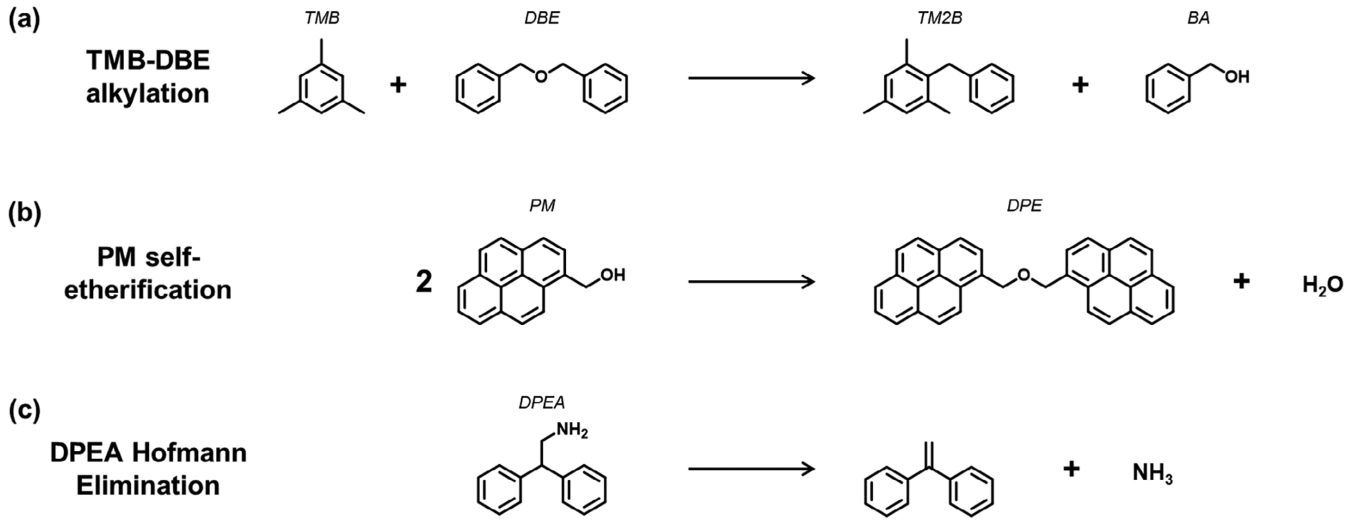
Mass-normalized SBAS densities in EFAl-MFI zeolites decreased relative to their parent catalysts, but to a much greater extent than total BAS densities. Following steaming total BAS density in H-MFI-25 was reduced by 11%, but SBAS density fell by an entire order of magnitude, suggesting that surface framework aluminum atoms are more easily converted to EFAl than microporous ones. A similar phenomenon was observed in EFAl-H-MFI-40 (52% decrease in SBAS density versus parent), but to a lower extent than EFAl-H-MFI-25, possibly due to lower aluminum content. For AHFS-MFI zeolites, changes in measured SBAS densities were consistent with how the AHFS treatments affected MFI mesopores. Mass-normalized SBAS density of AHFS-H-MFI-25 decreased by 51% relative to H-MFI-25, which was likely a combined effect of dealumination (via removal of surface framework aluminum species) and reductions in S_{meso} and V_{meso} (which limited access to mesopore-accessible BAS). In contrast, increases in S_{meso} and V_{meso} in AHFS-H-MFI-40 overcompensated concomitant framework aluminum losses resulting in an overall 72% increase in SBAS density.

SBAS densities extracted for parent and modified MFI zeolites spanned multiple orders of magnitude, which makes this catalyst suite appropriate to develop additional correlations between kinetic rate constants and SBAS counts for other probe reactions involving micropore-occluded molecules. The following section details extraction of SBAS counts for large-pore zeolites as enabled by extrapolation from reactions on MFI surfaces of known SBAS densities.

3.3 | Determination of SBAS Densities in Large-Pore Zeolites via PM Self-Etherification

SBAS densities in MFI zeolites could be determined via the TMB-DBE alkylation reaction due to the inability of TMB molecules to access microporous BAS, thereby relegating alkylation turnovers to SBAS. However, this reaction probe is not similarly applicable to large-pore zeolites because TMB is not diffusion-prohibited from accessing micropores within these frameworks [40, 44–46]. To account for the larger pore limiting diameters of 12-membered ring zeolites, batch self-etherification reactions were investigated utilizing 1-pyrenemethanol (PM; Scheme 1b), a polycyclic primary alcohol shown to be unable to diffuse into FAU micropores [23]. Notably, during PM self-etherification reactions, the alkylation product between PM and the TMB solvent (analogous to TM2B formed from BA and TMB) was not observed, likely indicating formation of the corresponding PM-TMB transition state was either energetically unfavorable or hindered sterically by the pyrene group.

Measured PM self-etherification rates were zero-order in PM, as evidenced by ether formation rates on H-MFI-40 and H-FAU-



SCHEME 1 | Reaction probes utilized in this study.

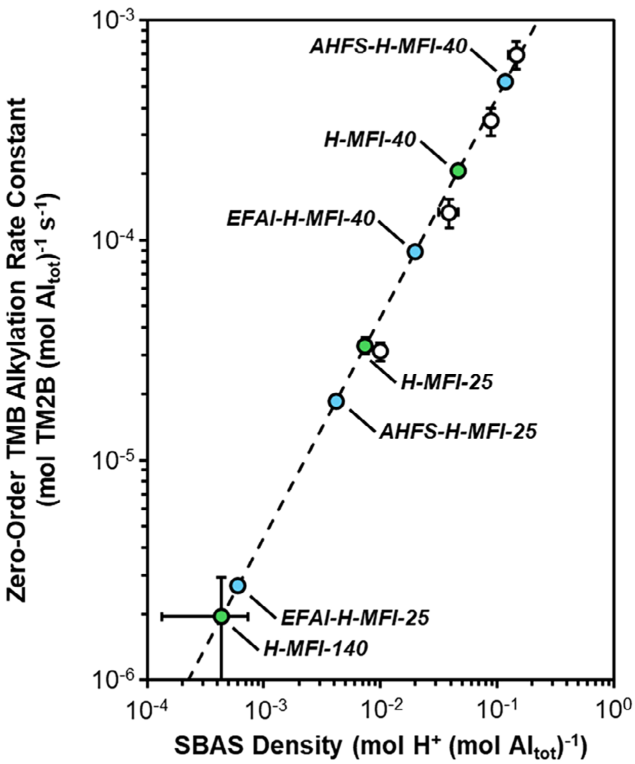


FIGURE 3 | Aluminum-normalized zero-order TMB-DBE alkylation rate constants as a function of SBAS densities for parent MFI zeolites (green) and steamed and AHFS-modified MFI (blue). The dotted line represents the linear correlation by Ezenwa et al. obtained by regressing MFI zeolites (white) through the origin [14]. Reaction conditions: 363 K, 5 cm^3 TMB, 0.2 cm^3 DBE, and 100 mg catalyst.

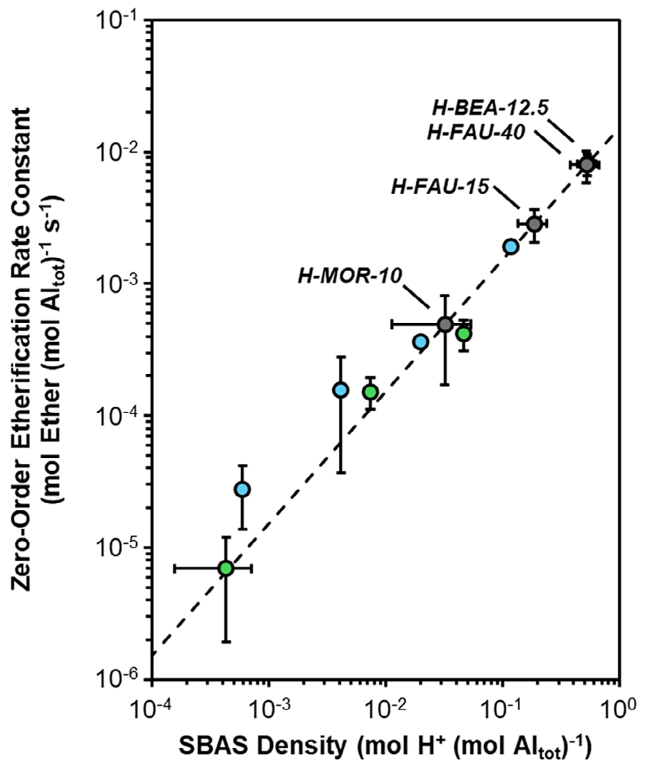


FIGURE 4 | Aluminum-normalized zero-order PM self-etherification rate constants as a function of SBAS for parent MFI zeolites (green), EFAl- and AHFS-modified MFI (blue), and large-pore zeolites (gray). Linear correlation represented by the dotted line was obtained by regressing parent, EFAl-, AHFS-modified MFI zeolites through the origin. Reaction conditions: 363 K, 10 cm^3 TMB, 100 mg PM, 3–50 mg catalyst.

40 that did not vary for initial PM concentrations between 20 and 40 mM (Figure S12). Zero-order PM self-etherification rates suggested the operative reaction mechanism proceeded via rate-limiting ether formation from co-adsorbed PM dimers on BAS, and adsorbed PM dimers were the most abundant surface species (Section S2). Measured aluminum-normalized self-etherification rate constants (k_{ether}) were plotted against aluminum-normalized SBAS densities extracted from TMB-DBE alkylation reactions for parent and modified MFI zeolites (Figure 4).

Plotting k_{ether} against SBAS density resulted in a linear trend with strong positive correlation (Pearson's r = 0.98) and high goodness of fit (R^2 = 0.97) with a slope of 0.015 ± 0.002 mol ether

TABLE 3 | Measured aluminum-normalized zero-order PM self-etherification rate constants and corresponding SBAS densities extracted via the linear correlation given in Figure 4. Values in parentheses represent 95% confidence interval uncertainties.

Catalyst	k_{ether} (mmol ether (mol Al_{tot})$^{-1}$ s^{-1})		SBAS density (mmol H^+ (mol Al_{tot})$^{-1}$)		SBAS density (mmol H^+ g_{cat}^{-1})	
H-MFI-25	0.15	(0.04)	10	(3)	0.0052	(0.0021)
H-MFI-40	0.42	(0.11)	27	(7)	0.011	(0.004)
H-MFI-140	0.007	(0.005)	0.5	(0.3)	0.00005	(0.00003)
EFAl-H-MFI-25	0.028	(0.014)	1.8	(0.9)	0.0009	(0.0005)
EFAl-H-MFI-40	0.36	(0.02)	24	(1)	0.010	(0.001)
AHFS-H-MFI-25	0.16	(0.12)	10	(8)	0.0046	(0.0036)
AHFS-H-MFI-40	1.9	(0.1)	120	(10)	0.033	(0.002)
H-MOR-10	0.49	(0.32)	32	(21)	0.052	(0.031)
H-BEA-12.5	8.1	(1.5)	530	(100)	0.55	(0.09)
H-FAU-15	2.9	(0.8)	190	(50)	0.17	(0.05)
H-FAU-40	8.0	(2.2)	520	(140)	0.13	(0.03)

(mol H^+ s)$^{-1}$ representative of the intrinsic SBAS-normalized zero-order rate constant. The linear plot spanned over two orders of magnitude in SBAS, consistent with findings that acid strength in MFI zeolites does not depend on concentration of isolated BAS [47]. Further, this linear correlation demonstrated that, identical to the TMB-DBE alkylation system, PM self-etherification rates solely depended on the number of SBAS accessible for the reaction. The sole dependence on SBAS was also consistent with the linear regression intercepting the origin (Figure 4), indicating that no reaction would be expected in the absence of SBAS. SBAS-dependent rates were further supported by steamed and AHFS-treated MFI zeolites with varying EFAl content (and therefore, varying Lewis acid site densities) versus their parent catalysts that collapsed to the linear regression, suggesting that Lewis acids did not catalyze PM self-etherification, or at least that they were scarce enough to contribute negligibly to rates. Finally, no reactivity for PM self-etherification was measured on aluminum-free silicalite-1 (Figure S13), indicating that silanols did not catalyze self-etherification turnovers.

This linear relationship between k_{ether} and SBAS density in MFI zeolites provided a means to determine SBAS densities in large-pore zeolites via calculation and mapping of their k_{ether} values. Extension to MOR, BEA, and FAU frameworks required the assumption of equal Brønsted acid strength to MFI, but work by Jones and Iglesia has demonstrated that deprotonation energies are independent of zeolite framework type (e.g., MFI, MOR, BEA, FAU) and T-site location when rigorously accounting for confinement effects [48]. In addition to BAS not differing in acid strength across zeolite frameworks, it is also emphasized that these reactions experience minimal confinement effects due to transformations occurring on SBAS, and that this data is consistent with Brønsted acid strength of SBAS and internal BAS that are similar [49, 50]. Therefore, extension of the correlation given by MFI zeolites to calculate SBAS densities in large-pore zeolites (Table 3) was valid.

Among large-pore frameworks, H-BEA-12.5 had the highest measured mass-normalized SBAS density (0.55 ± 0.09 mmol H^+ g_{cat}^{-1}) consistent with its high total aluminum content and exceedingly small crystallite diameters (~20–30 nm, Figure S4) resulting in a naturally high external proton density. SBAS densities of H-FAU-15 (0.17 ± 0.05 mmol H^+ g_{cat}^{-1}) and H-FAU-40 (0.13 ± 0.03 mmol H^+ g_{cat}^{-1}) exceeded that of H-MOR-10 (0.052 ± 0.031 mmol H^+ g_{cat}^{-1}) despite similar crystallite diameters across the three zeolites samples. These high relative SBAS densities in the FAU zeolites could reflect increased BAS access by mesopores generated via strong dealumination via steaming to raise Si/Al ratios from low values typical of FAU syntheses [51] to the Si/Al = 15, 40 range. Indeed, mesopores in H-FAU-15 and H-FAU-40 were evidenced by N_2 physisorption hysteresis behavior (Figure S1), BJH adsorption pore size distributions (Figure S2), TEM imaging (Figure S4), and S_{meso} values that exceeded that of H-MOR-10 by over a factor of five (Table 1).

These results taken together present PM self-etherification as a powerful tool to quantify SBAS densities in large-pore zeolite frameworks due to the sole dependence of di-(1-pyrenylmethyl)ether formation rates on SBAS. SBAS densities extracted for MOR, BEA, and FAU zeolites were reasonable when contextualized with their material properties, suggesting that inherent per-SBAS rates on these frameworks were identical to those on MFI zeolites. Potentially, this may experimentally indicate that zeolitic BAS demonstrate equal reactivity irrespective of framework type when protons reside on external crystal surfaces, at least in part because pore confinement effects are minimal. This hypothesis is probed further in the following section where SBAS densities are assessed as descriptors for reactions involving bulky molecules on zeolite surfaces that depend on external proton counts.

3.4 | Implications of SBAS Density for Reactions on Zeolite Surfaces

Evaluation of SBAS density is particularly attractive for its potential to explain reaction behavior when catalytic turnovers on external surfaces are relevant due to diffusion limitation

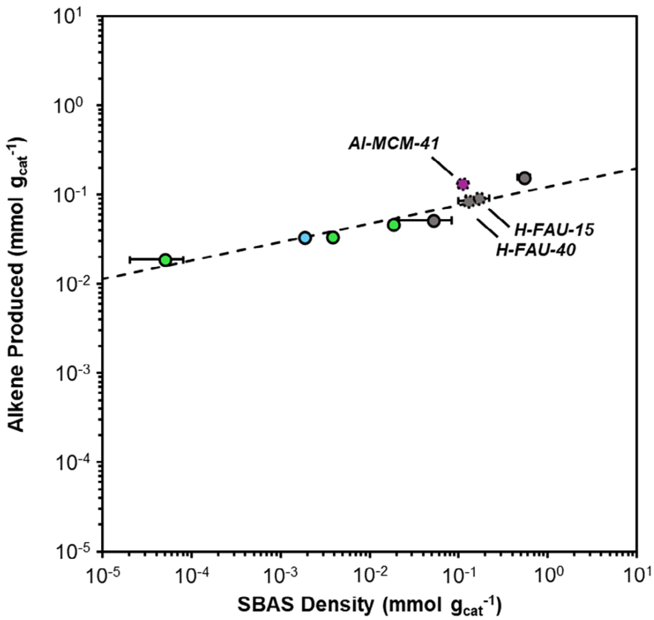


FIGURE 5 | Mass-normalized alkene production values in DPEA Hofmann elimination reactions plotted against SBAS densities for MFI (green), AHFS-H-MFI-25 (blue), large-pore zeolites (gray), and Al-MCM-41 (purple). SBAS densities are determined by TMB-DBE alkylation (parent, AHFS-MFI, Al-MCM-41), or PM self-etherification (large-pore zeolites). Denoted by dotted data points for H-FAU-15, H-FAU-40, and Al-MCM-41, these alkene production values were adjusted to account for the presence of mesopores (Section S3). The linear fit was regressed through all samples except H-FAU-15, H-FAU-40, and Al-MCM-41.

or inaccessibility of reactant molecules to microporous BAS (which often exceed SBAS by ≥2 orders of magnitude). To minimize possible solvent effects present in liquid-phase TMB-DBE alkylation and PM self-etherification reactions during SBAS quantification, Hofmann elimination reactions were performed on 2,2-diphenylethylamine (DPEA; Scheme 1c) shown by Pereira and Gorte to be inaccessible to microporous BAS in thermogravimetric studies on MFI and FAU zeolite [22]. Mass-normalized alkene production values (confirmed by TGA-MS analysis, Figure S14) were compared to corresponding mass-normalized SBAS densities for medium- and large-pore zeolites (Figure 5).

Notably, mass-normalized alkene productions exceeded mass-normalized SBAS densities by orders of magnitude in some samples (e.g., H-MFI-140) as demonstrated by deviation of the catalyst suite from a 45 ° slope which would imply perfect parity. This deviation indicated that the washing procedure utilized following DPEA exposure failed to completely remove residual weakly adsorbed (i.e., physisorbed or adsorbed to Lewis acid sites) DPEA. Despite this, a strong, positive linear correlation (Pearson's $r = 0.95$), between alkene production and SBAS density with high goodness of fit ($R^2 = 0.89$) was observed for H-MFI-25, H-MFI-40, H-MFI-140, AHFS-H-MFI-25, H-MOR-10, and H-BEA-12.5. This strong linear trend indicated that while dosed DPEA molecules exceeded SBAS counts, DPEA:SBAS ratios were in excess of 1:1 to a similar extent likely due to comparable V_{meso} values (0.04–0.11 cm^3 g_{cat}^{-1}) for parent, EFAl-, and AHFS-MFI zeolites (Table 1). Among the microporous zeolites, H-BEA-12.5 deviated the most from linearity (+50% higher alkene production versus expected), but this increase was likely due to particularly small crystallites (Figure S4) that yielded a high external surface area to which a greater amount of excess DPEA adsorbed relative to MFI and MOR. Indeed, H-BEA-12.5 exhibited high apparent S_{meso} (230 m^2 g_{cat}^{-1}) and V_{meso} (0.78 cm^3 g_{cat}^{-1}) despite absence of a type H4 hysteresis loop characteristic of mesoporous zeolites in its N_2 physisorption isotherm [52], which was present in MFI and FAU zeolites (Figure S1).

In contrast to MFI, MOR, and BEA zeolites, H-FAU-15 and H-FAU-40 exhibited significant positive deviations from the linear correlation (~151% and ~119%; Figure S16) likely suggesting that a greater relative amount of excess DPEA was present due to their inherent mesopores. To assess whether mesopores could explain greater alkene production in FAU zeolites, DPEA Hofmann elimination was performed with Al-MCM-41, a purely mesoporous Brønsted acid aluminosilicate catalyst. Consistent with FAU zeolites, alkene production in Al-MCM-41 was >400% higher than estimated by the linear correlation at that given SBAS density (Figure S16). Further, the larger deviation of Al-MCM-41 from linear relative to FAU zeolites was sensible considering S_{meso} and V_{meso} values exceeded those of FAU by a factor of ~4 (Table 1).

These results suggested that increased mesoporosity was indeed related to greater alkene productions in FAU and Al-MCM-41 relative to the primarily microporous MFI, MOR, and BEA catalysts. To deconvolute the effects of mesopores and SBAS density on DPEA Hofmann elimination reactions, alkene production rates by Na^+-exchanged FAU and Al-MCM-41 were compared to their Brønsted acidic counterparts. Although sodium exchange procedures did not exchange all surface protons with Na^+ counterions (29%–99% exchanged; Table S1), these materials presented as ideal reference catalysts at lower SBAS densities but identical S_{meso}, V_{meso}, and mesopore architectures from which contributions of mesopores on alkene production were estimated (Section S3). When accounting for (and subtracting the effects of) mesopores on alkene production, H-FAU-15 and H-FAU-40 collapsed to the linear correlation comprised by MFI, MOR, and BEA zeolites (+6% and +5% deviation from linear), but a +68% disparity in alkene production remained in Al-MCM-41 (Figure 5). Although a non-negligible deviation from linear persisted in Al-MCM-41, a further collapse to linearity was not necessarily expected due to differences in mesopore dimensionality. While H-FAU-15 and H-FAU-40 contained random 3D mesopores that enabled relatively facile egress of DPEA molecules (i.e., more similar to diffusion from external surfaces), the ordered 1D mesopores of Al-MCM-41 begat longer intracrystalline residence times that increased the likelihood of additional turnovers events occurring during DPEA elution. Therefore, higher alkene production in Al-MCM-41 relative to those on external surfaces or within 3D mesopores of zeolite catalysts was reasonable.

Strong linear correlation between alkene production and SBAS density in DPEA Hofmann elimination reactions suggested similar access of this reaction to TMB-DBE alkylation and PM self-etherification reactions utilized to extract SBAS values. Near-unity Pearson's r and R^2 values (0.95, 0.89) for the linear fit also indicated that extracted SBAS values are quantitative and, consequently, that PM self-etherification is a valid reaction probe to measure SBAS densities in these materials. Having established that measured SBAS densities in large-pore zeolites

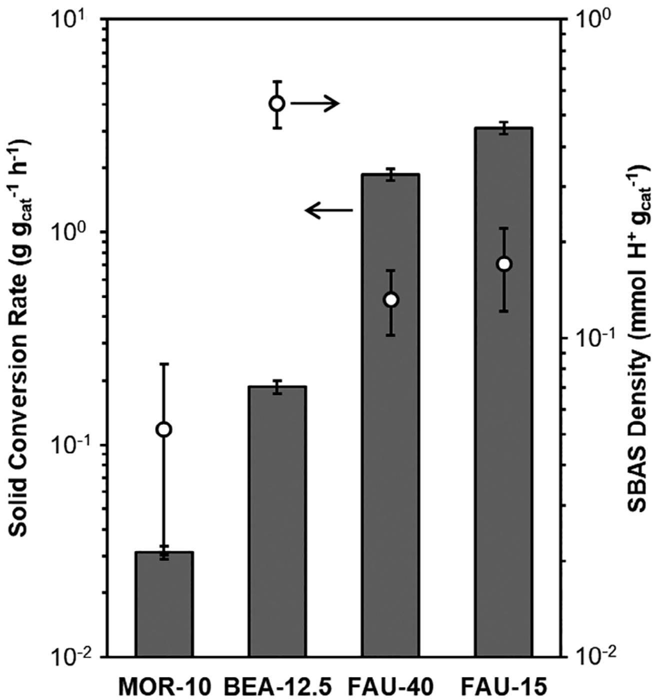


FIGURE 6 | Mass-normalized solid conversion rates at 5% conversion in PE upcycling reactions and mass-normalized SBAS densities determined by PM self-etherification for proton form large-pore zeolites. Reaction conditions: 523 K, 30 cm^3 min^{-1} Ar, 5:1 PE:g_{cat}.

were quantitatively sound, SBAS density was evaluated as an activity descriptor for catalytic cracking of bulky PE molecules.

PE deconstruction by zeolite catalysts at mild temperatures (e.g., 473–573 K) has been suggested to be reliant on the presence of BAS on external crystallite surfaces to catalyze initial activation events of extremely diffusion-limited polymer chains prior to subsequent rapid secondary reactions facilitated by internal protons. Previous work in our group found that PE conversion rates on MFI approached rates of catalyst-free thermal PE degradation when SBAS were titrated with 1,3,5-trimethylpyridine, demonstrating that SBAS are necessary for PE conversion on purely microporous zeolites [25]. Work by Duan et al. similarly showed that enhanced C_1-C_7 product yield and reduced coking in MFI nanosheets relative to MFI nanocrystals were resultant of high external surface area of MFI nanosheets (and therefore, high SBAS density) that facilitated rapid transport of activated intermediates into micropores [53]. Others have similarly linked higher polyolefin catalytic cracking activity to greater SBAS counts though series of catalysts of varying BAS densities, SBAS titrations, or growth of inert external siliceous shells [18, 54, 55]. This implicit dependence of PE conversion rates on SBAS densities makes PE catalytic cracking an appropriate reaction to assess how SBAS densities extracted by PM self-etherification in MOR, BEA, and FAU zeolites impact catalyst activity when microporous BAS are inaccessible to reactant molecules.

Solid conversion rates at iso-differential conversion (5% by mass) for H-MOR-10, H-BEA-12.5, H-FAU-15, and H-FAU-40 were compared to their respective SBAS densities in PE catalytic cracking reactions (Figure 6). H-MOR-10 recorded the lowest solid conversion rate (0.031 ± 0.002 g g_{cat}^{-1} h^{-1}) among large-pore zeolites, consistent with having the lowest SBAS density (0.052 ± 0.031 mmol H^+ g_{cat}^{-1}) and 1-dimensional micropores that rendered slow ingress of reactants/intermediates and egress of products relative to the 3D micropores of BEA and FAU. Notably, despite having the highest measured SBAS density (0.55 ± 0.09 mmol H^+ g_{cat}^{-1}) H-BEA-12.5 had only the next-highest solid conversion rate (0.19 ± 0.01 g g_{cat}^{-1} h^{-1}). Comparison of this solid conversion rate in BEA with that of MOR was valid since both zeolites were essentially purely microporous as indicated by hysteresis behaviors in N_2 physisorption (Figure S1), and their relative solid conversion rates (BEA > MOR) reflected their associated SBAS densities (BEA > MOR). In the case of FAU, however, solid conversion rates in H-FAU-15 and H-FAU-40 exceeded that of H-BEA-12.5 despite higher SBAS densities in the latter. Nonetheless, this apparent discrepancy is rectified when considering the corresponding pore topologies. While it is hypothesized that PE catalytic cracking reactions are rate-limited by activation of polymer chains on SBAS, activated intermediates must also undergo subsequent intracrystalline diffusion for scission events to occur at substantial rates. As such, reported conversion rates of bulky PE molecules are not necessarily kinetic in origin, rather they may reflect ingress rates of cleaved intermediates, or even external diffusion of the polymer to SBAS. It follows that PE activation was faster in H-BEA-12.5 owing to a higher SBAS density than H-FAU-15 and H-FAU-40, but diffusion of the activated intermediates was likely slower in BEA since FAU zeolites are mesoporous. Therefore, compensation of SBAS densities a factor of ~3 lower in FAU by effective diffusivities higher than BEA plausibly yielded higher solid conversion rates in FAU zeolites. Finally, in comparing H-FAU-15 to H-FAU-40, a 66% higher solid conversion rate was measured in H-FAU-15 in agreement with higher SBAS density (0.17 versus 0.13 mmol H^+ g_{cat}^{-1}), a comparison enabled by similar S_{meso} (100 versus 110 m^2 g_{cat}^{-1}) and V_{meso} (0.18 versus 0.20 cm^3 g_{cat}^{-1}) values.

4 | Conclusions

Surface Brønsted acid site densities (SBAS) in large-pore zeolites MOR, BEA, and FAU were quantified via reactions utilizing micropore-occluded molecules that were catalyzed exclusively by sites located on external crystallite surfaces. A linear correlation between SBAS density and zero-order rate constants for self-etherification of 1-pyrenemethanol (PM) was developed using a suite of MFI zeolites post-synthetically modified via steaming and ammonium hexafluorosilicate treatments to yield catalysts with SBAS values as determined by surface alkylation reactions that varied by multiple orders of magnitude. The linearity of this correlation indicated that PM self-etherification rates solely depended on the number of SBAS available for the reaction, which therefore provided a basis to determine SBAS densities in large-pore zeolites by mapping of their respective self-etherification rate constants. SBAS values determined for MFI, MOR, BEA, and FAU zeolites were reasonable when contextualized with their textural properties. Further, 2,2-diphenylethylamine Hofmann elimination reactions on external surfaces trended with SBAS densities across all zeolites when accounting for mesoporosity in FAU. PE catalytic cracking reactions found to hinge on activation by SBAS and subsequent intracrystalline diffusion further highlighted the ability of SBAS quantification to elucidate reactivity trends and reaction pathways in systems under extreme diffusion limitation. The findings and associated catalytic methodology of this work can be applied generally to other systems involving diffusion-limited or pore-occluded molecules to quantitatively assess the pronounced catalytic consequences of SBAS under such conditions.

Acknowledgments

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data supporting the results of this work are available in the Supporting Information and upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

The Supporting Information contains additional catalyst characterization including porosimetry, electron microscopy, and raw data including sample ^{1}H NMR spectra, TMB-DBE alkylation, and PM self-etherification product concentration profiles, thermogravimetric PE catalytic cracking plots, and mechanistic discussion for PM self-etherification and DPEA Hofmann elimination reactions. Additional publications not referenced in the main text are also cited [56, 57].

Supporting File: cctc70613-sup-0001-SuppMat.docx.